Fake News Detection Using Deep Learning

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Course Title
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Introduction

Problem Statement

Fake news threatens personal, organisational and institutions by creating distrust and damaging reputations. Fake news can be particularly damaging as even if it can be refuted, it may still cause lasting damage to reputations. With the widespread use of social media, the proliferation of fake news has greatly increased both in terms of speed and quantity. This has led to the sheer volume of fake news (and real news) overwhelming human fact-checkers and preventing them from manually verifying all content.

But what is fake news? According to Baptista and Gradim (2022) the definition of fake news is generally agreed to be either completely or partially false information, with varying levels of agreement for other additional criteria. Such as the intent to deceive, and if only content in a news format should be counted. For the purpose of this research, the working definition of fake news defined by Baptista and Gradim (2022) will be used:

> a type of online disinformation (1), with (2) misleading and/or false statements that may or may not be associated with real events, (3) intentionally created to mislead and/or manipulate a public (4) specific or imagined, (5) through the appearance of a news format with an opportunistic structure (title, image, content) to attract the reader's attention, in order to obtain more clicks and shares and, therefore, greater advertising revenue and/or ideological gain Baptista and Gradim (2022)

In a recent (as of writing) example of fake news causing adverse impact, Savage (2025) wrote for The Guardian about YouTube channels spreading fake news about UK politics in their videos, acruing close to 1.2 billion views in total. While some forms of news which are factually fake are usually harmless (parody, satire, etc.) fake news, with the intent or deceiving those who consume it is dangerous, it can widen division, spread prejudice and cause distrust in institutions are among just a few examples. And while fake news may sound innoculous enough, 'It's just fake news, ignore it', it can lead to serious consequnces, including injury and death. In one such example, during the COVID-19 vaccination campaign in Taiwan an adverse correlation between prevalence of fake news and vaccination rates were found (Chen et al., 2022).

As a result, the development of automated fake news detection systems have been explored. But limitations related to generalisation where detection systems perform poorly on data which did not appear in its training sets (Qin & Zhang, 2024) limit their applicability. Additionally, current methods such as machine learning approaches, fail to capture the semantic data meaningfully, reducing its detection accuracy. Furthermore, many approaches which employ deep learning are often black-boxes, which results in lowered trust towards these systems (Szczepański et al., 2021).

This highlights the need for new approaches which are capable of detecting fake news by analysing the semantic information in the text throughly, having greater generalisation and be more transparent (explainable) when making its classifications.

Research Objectives

The objectives of this project are:

- Review on methods to detect fake news
- Develop a fake news detection using deep learning
- Evaluate the performance of model in detection fake news

Research Scope

This project aims to build a deep learning model for fake news detection by first conducting a thorough literature review on related works in order to construct a knowledge base of which will be used to construct a model for detecting fake news.

After the literature review, a model to detect fake news will be developed using deep learning techniques. This process will involve gathering datasets, analysing and understanding the data and pre-processing datasets before constructing the architecture of the models and then training the various models.

And lastly, after training the models. The models will be tested on the test sets which are subsets of the datasets used in order to determine their performance.

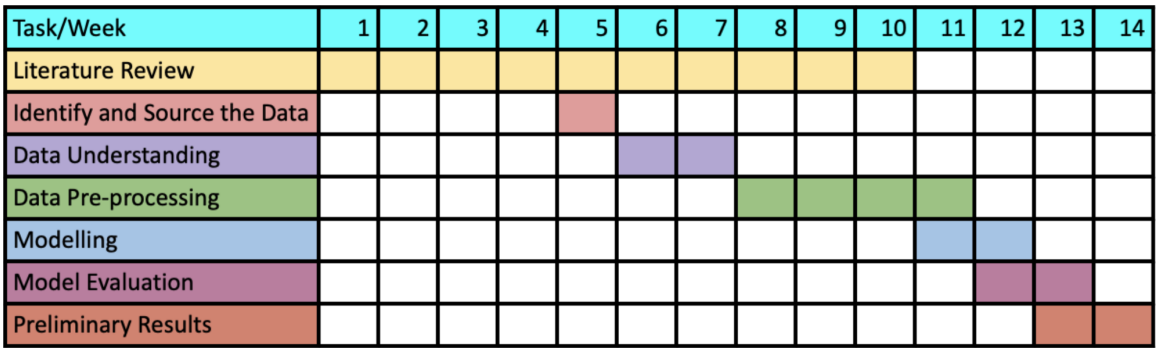

Figure 1

Gantt chart depicting the timeline for this project relative during MMU's term 2530

Literature Review

Deep Learning Methods and Other Modelling Techniques

Convolutional Neural Networks

In research conducted by (Q. Zhang et al., 2023), which detected fake news on Chinese social media networks using an augmented Convolutional Neural Network (CNN), Convolutional Feed-Forward or Conv-FFD. In order to process the Chinese characters, the researchers used a three-step process starting with one-hot-encoding the over 4000 Chinese characters, then feeding this data in an embedding process which converts the one-hot-encoded data into vector representations of each character, culminating in iterating through all the characters to get the vector representation of each one (essentially mapping the characters to the vector embedding). As for the feature modelling process (the process of understanding the relationship between characters/words), the researchers used the CNN model which is able to learn hidden features from conducting convolution operations. They used a 'sliding window' in order to create sentence-level embedding. Then in order to classify the text as real or fake, a softmax activation function is used in order to predict the class of the input text. The model achieved an f1-score of 0.8702 - 0.8855, indicating it is capable of predicting whether a given body of (Chinese) text is real or fake.

Research conducted by (Saleh et al., 2021) also proposed a CNN model, albeit a modified one dubbed "optimized Convolutional Neural Network to detect fake news" or OPCNN-FAKE. Aside from the proposed models, the authors also built machine learning models and other deep learning models like LSTMs The authors used four datasets, Fake News Detection (from Kaggle), FakeNewsNet, FA-KES5, and the ISOT dataset. The datasets were cleaned by converting all characters to lowercase, removing URLS, removing special symbols, removing stop words, tokenisation and stemming (reduction to root word). After the data was cleaned, two methodologies were used for feature generation, one for training the machine learning models and one for the deep learning models. For machine learning models, the feature generation used two techniques, N-grams (up to n = 4) and Term Frequency - Inverse Document Frequency (TD-IDF). While for the deep learning models, feature generation was done via word embeddings (the authors classified this as the "Embedding Layer" or the

first layer of the proposed OPCNN-FAKE model), specifically the Global Vectors for Word Representations (GLoVe) which converted the cleaned textual data into dense vectors (more specifically they used glove6B.zip). The preprocessed datasets were then split 80-20 into training and test sets respectively and the training set was used to train the machine learning and deep learning models, including the proposed OPCNN-FAKE model. For the deep learning models, a technique called "Hyperopt Optimisation" was used to adapt the various model parameters including epoch, dropout and batch size, among others. The proposed OPCNN-FAKE model was configured with six layers, the embedding layer, dropout layer, convolutional layer, pooling layer, flatten layer and output layer respectively. Without diving too deeply into the architecture, the embedding layer was previously explained; the dropout layer randomly deactivates some neurons to mitigate overfitting; the convolutional layer uses a convolution filter and feature map to identify the data patterns and high and low level features while employing a ReLu activation function; the pooling layer reduces the number of features in the feature map so as to only retain the most important features; the flatten layer flattens the feature map into a one-dimensional array; the output layer takes the one dimensional array from the flatten layer and classifies it as fake (0) or true (1). The models were then evaluated using cross-validation and the OPCNN-FAKE model achieved the best f1-score across all four datasets used, with the proposed model achieving an f1-score of 0.959 on the Kaggle dataset and a perfect f1-score of 0.9999 on the ISOT dataset.

Hybrid Approaches

There has also been research that looked into hybrid approaches, (Hashmi et al., 2024) proposed a hybrid deep learning model combining Long Short Term Memory (LSTM) and CNN. Using publicly available datasets (WELFake, FakeNewsNet and FakeNewsPrediction), the authors first pre-processed the data using lemmatisation (reduction of words to root form), de-capitalisation, tokenisation, among others. To process the cleaned data, the authors used FastText, both its supervised and unsupervised forms to create word embeddings. These words embeddings, generated from both the unspervised and supervised FastText models were then used to train separate models, including ensemble (stacked) machine learning models, transformer models and the CNN-LSTM hybrid mentioned earlier. Evaluation of the models using test subsets of each of the datasets used showed that those trained on unsupervised FastText embeddings performed worse than those trained on supervised FastText embeddings, with f1-scores between 0.83 - 0.97 with significant variation between the datasets it was tested on. Those trained on supervised FastText embeddings were evaluated on subsets of the datasets used (test sets) and had f1-scores of between 0.91 - 0.99. The CNN-LSTM model performed better than all other models (stacked machine learning models and transformer models), with high f1-scores of between 0.97 and 0.99 across all datasets.

While most research seems to have been done on detecting fake news in English, research conducted by E.Almandouh et al. (2024) detected fake news in Arabic. The authors used two datasets, Arabic Fake News Dataset (AFND) and their own dataset scraped from Twitter. The authors explored multiple types of models for detecting fake news including traditional machine learning and common deep learning models but they also proposed a hybrid model called Bidirectional Long Short Term Memory – Bidirectional Gated Recurrent Unit (Bi-LSTM – Bi-GRU). This proposed model is structured sequentially, but it can also deal with information bi-directionally and it attempts to leverage the strengths of Bi-LSTMS and Bi-GRUs. Bi-LSTMs and LSTMs in general are excellent at retaining information and addresses the vanishing gradient problem; while GRUs are able to mamnage shorter dependencies and are computationally efficient. For preprocessing, the authors first cleaned the data by removing all punctuation, unicode characters, names, URLs, stop word removal, tokenisation and other standard techniques. The cleaned data was then features were extracted using both supervised and unsupervised FastText which were then used to train models. The Bi-LSTM – Bi-GRU performed the best among all of the models tested, achieving f1-scores of between 0.98 and 0.99 depending on the dataset.

(Dhiman et al., 2024) proposed a new deep learning model for predicting fake news called "Generative Bidirectional Encoder Representations from Transformers" or GBERT, a hybrid of Generative Pre-Trained Transformer (GPT) and Bidirection Encoder Representations from Transformers (BERT). The authors used two benchmark Indian datasets: The Indian Fake News Dataset (IFND) and FakeNewsIndia Dataset. The standard

cleaning and normalisation of the textual data was applied including: decapitalisation, removal of URLs, names, punctuation stop words, numbers and emojis, and the application of stemming and lemmitisation. Now for the feature extraction the cleaned data is fed into two Large Language Model (LLM) components, BERT and GPT. BERT captures word embeddings, with context from every token inputted and outputs a summary of the context inside the input sequence. While GPT captures the global dependencies and semantic coherence in the input data and outputs the last hidden state of the final token. Finally the output from BERT and GPT (the features) are merged together in a concantenation layer before being passed to the dense layer for the final binary classification using a Sigmoid function. The results showed that the proposed GBERT model performed well, with a f1-score of 0.9623. It is worth noting, however, that XGBoost performed well in terms of accuracy (f1-score was not provided for XGBoost) with a score of 93.42% while the proposed GBERT model has a score of 95.30%.

Deep Ensemble Approaches

Other than approaches with CNNs, LSTMs and traditional machine learning models, there have also been attempts to construct deep ensemble models composed of deep learning models arranged in an ensemble architecture. Research conducted by (Ali et al., 2022) proposed a new approach for detecting fake news, by using Sequential deep learning (SDL) models as the base models in an ensemble stacking architecture with a multilevel perceptron classifier at the top. The authors used two datasets: the LIAR and ISOT datasets. The data from each dataset was preprocessed with tokenisation, lemmatisation among other standard techniques for working with textual data. Unlike with other similar works, the authors chose to do multi-class classification because fake news is usually mixed with true news. For the SDLs they are composed of dense and dropout layers. The ReLu activation function is used in all dense layers except the final output layer where a Sigmoid function was used. The ensemble model was evaluated with the test subset of each dataset used, for the LIAR dataset the model achieved f1-scores in a range from 0.48 to 0.82 for the various classes. As for the ISOT dataset, since the dataset only had two classes (real and fake), the model only had to classify true and false, the binary classification was able to achieve a 1.00 accuracy using the 2nd stage multi-class classification.

Machine Learning Approaches

Research conducted by (Khanam et al., 2021) explored the use of machine learning models such as XGBoost, Support Vector Machine (SVM), Naive Bayes, among others. The authors used the LIAR dataset which they then preprocessed by removing noise such as identification (ID), dots, commas and questions as well as stemming (reduction of a word to root form) and removing any suffixes. They then use Part of Speech (POS) to convert the text into tokens and values. They then extracted features by extracting lexical features such as word count, sections of speech etc. and then used the Term Frequency-Inverse Document Frequency (TD-IDF) vectorizer function from the publicly available library sklearn (also known as scikit-learn) to extract unigram and bigram features. The results showed that the accuracy of the models was limited, especially in comparison to approaches using deep learning with the best performing model being the XGBoost model with an accuracy of 0.73.

Multimodal Approaches

In research conducted by Jing et al. (2023) the researchers employed a multimodal approach instead of a purely textual approach to detection. To this end, the authors proposed the novel, end to end Multimodal Progressive Fusion Approach or MPFN. The authors used two datasets, a Twitter dataset and a Weibo dataset. The MPFN model has three components, a text feature extractor, visual feature extractor and the progressive multimodal feature fusion process. Starting with the text feature extractor, the authors used a pre-trained Bidirectional Encoder Representations from Transformers (BERT) in order to extract the features from the text, this BERT model returned text features with 768 dimensions. The visual feature extractor has two major subcomponents, the first seeking to obtain spatial semantic information and the second seeking to obtain features which can indicate whether an image has been tampered with. The first component starts off by setting all images to 224 x 224 x 3 (width, height, colour), then to extract features the authors used a pre-trained Swin

Transformer (SwinT) model to extract the spatial semantic information of the image. This is done by splitting the images into 4 x 4 x 4 non-overlapping patches and each patch is passed through four layers in order to obtain a hierarchical representation. As for the second subcomponent, the authors extracted frequency domain features using Discrete Fourier Transform (DFT) and Visual Geometry Group 19 (VGG19). DFT was used to convert the image from the spatial domain to the frequency domain and the output is inputted into the VGG19 model which outputs deep semantic features; however in order to obtain a hierarchical data structure, the outputs from the 2nd, 4th, 8th and 16th convolutional layers in the VGG19 model is extracted for use in the fusion model later. The outputs from the text extractor, and the outputs from the visual feature extractor are then combined in a progressive fusion approach where the shallower layers in the hierarchical data structure are formed first and so on for deeper layers (there are 4 layers total). Mlp mixer was used to fuse feature information in a finer way and enhanced the relationships between modes. The final fusion features are then used for classification by a fully connected layer using a softmax function; additionally the model uses a cross-entropy loss function for calculating the difference between the true label and predicted probability. The results showed that this approach achieved an f1-score of 0.889 - 0.876 for both the Weibo and Twitter datasets for both classes with the exception being a 0.740 score achieved by the model when classifying real news on Twitter.

In research conducted by Xue et al. (2021), the authors proposed a novel 5-module model "Multimodal Consistency Neural Network" or MCNN for multimodal (image, text) fake news detection as depicted in figure 3. The text feature extraction model uses a BERT pre-trained model to encode text and then uses Bidirection Gated Recurrent Unit (BiGRU) to extract temporal features and convert features into a sequence that is suitable for integration with image features. For the visual semantic feature extraction module, a pre-trained ResNet50 model is used to encode the image which is then fed into an attention mechanism to highlight emotional regions of the image followed by the use of BiGRU to convert the features into a visual semantic sequence. The visual tampering feature extraction model, unlike the previously described module seeks to extract features at a physical level in order to detect malicious tampering with the image, it uses the Error level Analysis (ELA) algorithm to process input images before using ResNet50 to extract features. The similarity measurement module maps the text and image semantic representations into one space, followed by the calculation of the cosine similarity; this is done to address the issue of image-text mismatch. This is finally followed by the multimodal fusion module which takes input from the similarity measurement module and the visual tampering feature extraction module and uses an attention mechanism to give the combined features weights before using a softmax classifier to classify the image. The model was tested against various baseline models, both single and multimodal models were used. Four distinct datasets were used and the model performed the best in terms of f1-score across all datasets save one, losing out to a BERT-MVNN hybrid (note: there appears to be an error in the paper where the authors failed to notice that the BERT-MVNN model outperformed the proposed model in the results table). The proposed model’s f1 score ranged from 0.968 - 0.831.

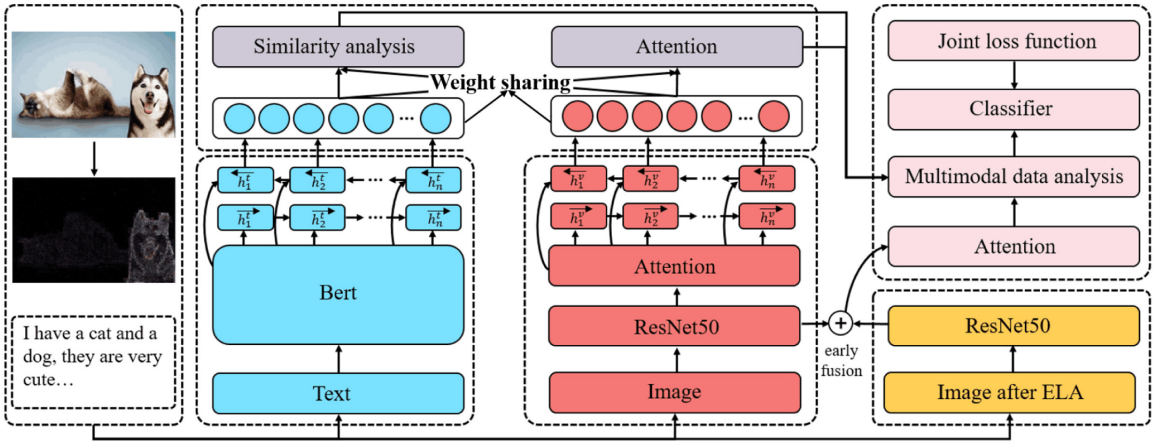

Figure 2

Note. Figure depicts the architecture of the proposed MCNN model. The module coloured blue is the text feature extraction model; the module coloured red is the visual semantic feature extraction module; the module coloured in purple (above the text and visual semantic feature extraction modules) is the similarity measurement module; the module in yellow/orange is the visual tampering feature extraction module; and lastly the module in pink is the multimodal fusion module. From "Detecting fake news by exploring the consistency of multimodal data", by Xue et al., 2021, Information Processing and Management., 58(5), https://doi.org/10.1016/j.ipm.2021.102610. Copyright 2024 held by 2021 Elsevier Ltd.

(Yuan et al., 2021) proposed a new model for detecting fake news, the "Domain-Adversarial & Graph Attention Neural Network" or DAGA-NN. This approach is multimodal and takes both image and text as input. The proposed model was designed in order to address the challenge machine learning approaches faced when attempting to classify cross-domain news samples that did not (or had few instances of) appear in its training data. The architecture is depicted in figure 4. Starting with the feature extractor module, the text component is first cleaned and tokenised before being passed to a BERT model to gain vector representations which are then passed to the Bi-LSTM to capture information about the text and temporal features. As for the image feature extraction, the input image is fed to a pre-trained VGG19 model and the output is passed to a fully connected neural network to gain the feature representation of the image. The text and image features are then fused together to form a single representation of the image and text. The fused feature representation is then fed to the domain discriminator whose goal it is to determine the domain information of the input news. The model is designed in such a way that the domain discriminator is put in competition with the feature-extractor module, forcing the module to output features which are not (or at least minimally) domain-specific, which helps the classifier train on features not dependent on domain and increasing its cross-domain classification accuracy. The domain discriminator is a two-layer neural network which outputs a vector of k-dimensions representing its probability of belonging to each of the k domains. Moving on to the graph-attention-based-classifier, it first takes news articles and creates a graph of them. Edges are drawn between nodes if their cosine similarity is high. Once the graph is built, the graph attention layer learns the importance of each node's neighbours when classifying an input article. This is critical as during this phase, the graph attention layer will strengthen connections between nodes (articles) which are similar semantically and have the same veracity label (true or false) and vice versa (weaken connections if veracity label is opposing). This establishes clear boundaries between real and fake news, even if they are semantically similar. The DAGA-NN model was tested against two datasets, the Twitter and Weibo datasets and managed to outperform state-of-the-art models. The DAGA-NN model was able to achieve macro f1-scores of 0.8828 and 0.9522 for the Twiier and Weibo datasets respectively. The best baseline model the DAGA-NN model was tested against was the Event Adversarial Neural Network or EANN and it outperformed it by 0.0890 - 0.0112 in terms of macro f1-score.

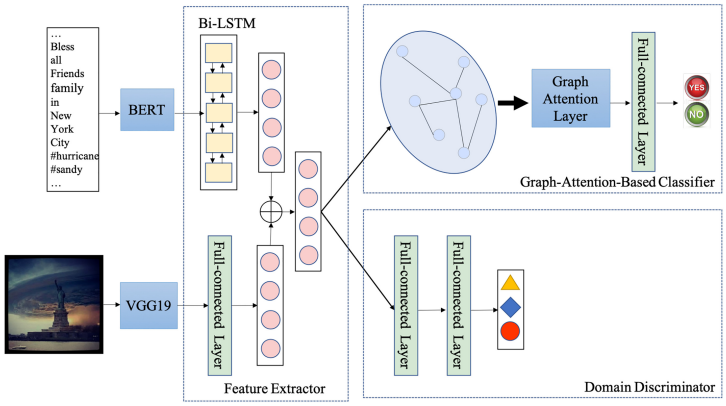

Figure 3

Note. Figure depicts the architecture of the DAGA-NN model, each box represents a module which is labelled in the buttom right corner. From "Improving fake news detection with domain-adversarial and graph-attention neural network" by Yuan et al., 2021, Decision Support Systems, 151, p. 3 (https://doi.org/10.1016/j.dss.2021.113633). Copyright 2021 Elsevier B.V.

Language Model Approaches

The rise of Large Language Models (LLMs) have complicated detection of fake news, to this end (Wu et al., 2024) proposed a new model "SheepDog" leveraging LLMs to improve detection. The model utilises two different models, an LLM for news re-framing and a fine-tuned language model for classification. Firstly, SheepDog utilises an LLM to transform news content into different styles so as to remove stylistic with varying prompts in order to transform a dataset into one which there is minimal stylistic differences between real and fake news. Then the LLM is further prompted with a set of questions to which the LLMs responses are used as pseudo-labels. Moving on to the language model stage, the language model is fine-tuned using labelled datasets processed by the LLMs. The language model will take input from the LLM stage and output the prediction (true or false news). SheepDog is designed to be able to work with any language model and as such the authors tested the SheepDog framework with several language models. For evaluation the authors tested SheepDog against LLM-empowered style attacks, unperturbed (original) articles, and adaptability with different language models. For the baseline models SheepDog was compared againts, the authors used three main groups of models: text-based fake news detectors, fine-tuned language models and LLMs. And the researchers used three datasets: PolitiFact, GossipCop and LUN. Sheepdog proved to be resilient against LLM-empowered style attacks and its performance was superior to all baseline models. When tested against unperturbed articles, SheepDog had comparable or superior performance relative to the baseline models. As for adaptability with various language models, the SheepDog framework was able to improve the performance of these language models. The best language model which was used as a backbone in the SheepDog framework depended on which dataset they were tested against but all language models integrated into the SheepDog framework performed better than its unintegrated counterpart.

Explainable Models

Research by Szczepański et al. (2021) proposed a novel approach which sought to make the model's decision making explainable (explainable AI), the reason being that having a black-box system determine the veracity of something doesn't inspire confidence, something an explainable AI model could mitigate. The researchers utilised DistilBERT which is a lighter variant of BERT, combined with an LSTM layer, a pooling layer, dense feed forward layer and finally the output layer. They then attached an explainability module to the black-box classifier. This module is composed of two post-hoc explainable AI techniques, Local Interpretable Model-Agnostic Explanations (LIME) and Anchors. Post-hoc refers to techniques which attempt to give some level of transparency to an otherwise black-box model. LIME works by sampling the model in the vicinity of a prediction (imagine having an unknown 'X' on a spot on a graph and then picking spots around it to approximate

the ’X”s coordinates) and the response to the sampling is used to train a linear model that approximates the model’s prediction that can be interpreted. While Anchors refer to an algorithm that applies ’rules’ to local predictions, the rules being ’if-then’ statements which should ensure that any prediction will remain the same wherever the rule holds. The model was able to achieve 0.98 f1-score for both positive and negative classes. The researchers were able to conclude from the results that it is feasible to use methods such as LIME and Anchors to explain the models behaviour. Although there was some overlap between LIME and the Anchors when explaining the model’s behaviour, the results showed it is possible to use multiple types of these methods can be used to derive explanations.

Chien et al. (2022) proposed an explainable AI approach, XFlag, to fake news detection which sought to increase the transparency and trust of black-box AI models. The authors used the Weibo dataset, which was processed to extract content, user and sentiment features. Content features were extracted with Doc2Vec which generates word embeddings from the content, similar to BERT, albeit Doc2Vec is an older model. User features were extracted from the user’s profile, such as followers, location, etc.. Sentiment features were extracted with the Google Cloud Natural Language API which returned the emotional, "sentimental" data from the content. These features are then passed to fully connected layers before being passed to the LSTM in order to standardise the input to the LSTM. XFlag was able to perform comparably to state of the art models, with an f1-score of 0.938. As for the explainability, the authors used the Layer-wise Relevance Propogation (LRP) algorithm which uses reverse propagation, essentially start with the model output and going in the reverse direction through the model’s various layers to determine which neurons had the most impact on the result. The output of the LRP would be the model’s (being tested) input shape but with the relevance score for each neuron, this is known as the explanation vector. LRP was applied to the XFlag model after the model was trained, and the output explanation vectors are then passed to Situation Awareness-based Agent Transparency (SAT) framework, specifically the SAT-3 (the researchers used other SAT versions from SAT 1 - 3 which each iteration having more insightful explanations, SAT-3 was most preferred by users) which takes this raw data and processes it in order to generate visualisations, textual annotations and the model’s uncertainty so that the user can see how the model came to the conclusion that it did. Users were tested on their trust of the XFlag model with various SAT versions, the lowest being SAT-1 which, similar to a model with no explainability, output the result without deeper explanation. The results were that users reported highest satisfaction and trust as the SAT level and correspondingly, the depth of the explanations of the model’s prediction, increased, the highest being SAT-3.

Research by Ahmed et al. (2022) explored the use of explainable AI (XAI) for detection of toxic COVID-19 fake news, proposing the Toxic Fake News Detection System (TFNDS) which uses Bidirectional Long Short Term Memory (BiLSTM) for classification and similar to Chien et al. (2022) the authors used LIME to explain the black-box model (the BiLSTM). The authors used two datasets which they subsequently merged together, the data was then preprocessed with removal of stop words, removal of emojis, removal of repetitive posts, lemmenisation and etc.. This data is then passed to a scikit-learn (also known as sklearn) pipeline where lists of words are indexed and turned into lists of indices and the length of this list is standardised by a transformer to a maximum length of 100. This is then passed to the model which is composed of 128 BiLSTM hidden units followed by 3 dense layers with a Sigmoid output layer for the final classification. As for the explainable section, LIME augments the data by removing features to see which features (words) was most important to the output and this information is visualised by highlighting the words. On the primary dataset used, the model was able to achieve an f1-score of 0.9425 and when tested on an additional dataset for verification (WNUT-2020) it achieved an f1-score of 89.40 which outperformed all other machine learning and deep learning models the authors built. The authors also sought to ensure the LIME explanations were correct, this was done by constructing a pipeline to find the Pointwise Mutual Information (PMI) of words throughout the entire dataset (in contrast to LIME which is local) and their association with a real or fake class. To conduct the evaluation, the 20 most important words for each post as found by LIME and extracted and compared to their rank in the PMI ranking using Kendall’s Tau correlation coefficient. The Kendall’s correlation for the BiLSTM was 0.35, the authors noted that the general range for Kendall’s Tau coefficient was between 0.3 and 0.4, indicating that the LIME explanations had issues with allignment as the PMI context is global while LIME is local.

Fine-tuning Approaches

An issue with current models is that while they work well with data which was found in their training sets, they often suffer in performance when they encounter data unknown to them (Qin & Zhang, 2024). To solve this issue, Qin and Zhang (2024) proposed a novel approach, Feature Regularization-based Adversarial Fine-tuning (FGM-FRAT). This approach generates perturbed versions of the input in order to confuse the model. This forces the model to find new, deeper patterns instead of relying on otherwise superficial features. The key innovation however, is the addition of a regularization term to the loss function which seeks to minimise the difference between the features generated from the unperturbed, original text with the features generated from the perturbed text. This ensures the feature representation generated by the model will remain similar between original and perturbed text, improving generalisation. The authors used the FakeNewsNet dataset to fine-tune a BERT model using the FGM-FRAT approach using the FakeNewsNet dataset. The authors then tested the fine-tuned model with unseen datasets (BuzzFeedNews, LIAR, WELFake and KaggleFakeNews), using the standard BERT model as a baseline as well as some other fine-tuned BERT models which were fine-tuned with various approaches. The BERT-FGM-FRAT model performed the best amongst all models tested, with improvements of up to 0.32 in terms of f1-score against the base BERT model on the BuzzFeedNews dataset. While for other datasets the BERT-FGM-FRAT managed to offer sigificant performance improvements with the sole exception of the WELFake dataset where it was only able to improve upon the base model incrementally with a 0.03 (f1-score) improvement.

Graph-based Approaches

Methods which seek to analyse propagation structure for fake news detection often use graph-based methods which overlook the semantic information being propagated, while methods which analyse the semantic information of the content usually fail to understand the propagation structure (Z. Zhang et al., 2024). Research by Z. Zhang et al. (2024) proposed a novel approach, Graph Convolution Network and BERT combined with Co-Attention (GBCA) which seeks to combine graph based analysis of propagation techniques while using BERT for semantic analysis in order to classify fake news. To capture the spread of news, the proposed model constructs two graphs to capture propagation and dispersation using Edged-enhanced Bayesian Graph Convolution Network (EBGCN). For semantic feature extraction, the GBCA uses a pre-trained BERT to generate vectors from the content which capture the semantic data. And finally to integrate the two approaches, the authors used a co-attention mechanism which computes an affinity matrix to find the relationship between the features from both the structural (propagation and dispersion) and semantic features, the co-attention mechanism will then dynamically assign weights to the features. The authors used three datasets, Twitter15, Twitter16, and PHEME; and used baseline methods including GRU-RNN, SVM-TK (Support Vector Machine - Tree Kernel) among other graph-based methods and deep learning methods. GBCA managed to outperform all models it was compared against in terms of accuracy with a score of 92.8%, 90.3% and 70.8% for the Twitter15, Twitter16, and PHEME datasets respectively; however it was beaten by several Graph Comvolution Networks (GCNs) when it came to f1-score for some of the classes. Despite this, it is worth noting the GBCA was more computationally efficient when training as the co-attention module used reduced the amount of iterations required for convergence.

Summary

Author(s)	Year	Title	Findings	Model Proposed	Additional Notes
Smith et al.	2020	Title of Paper 1	Key finding one	Proposed Model A	Note about methodology
Jones	2019	Title of Paper 2	Key finding two	Proposed Model B	Limitations noted
Miller, K.	2022	Title of Paper 3	Key finding three	Proposed Model C	Future work suggestions
Brown & Davis	2021	Title of Paper 4	Key finding four	Proposed Model D	Data source information
Garcia, J.	2018	Title of Paper 5	Key finding five	Proposed Model E	Conflicting result
Williams et al.	2023	Title of Paper 6	Key finding six	Proposed Model F	Replication study
Chen, L.	2020	Title of Paper 7	Key finding seven	Proposed Model G	Sample size details
Rodriguez	2017	Title of Paper 8	Key finding eight	Proposed Model H	Theoretical contribution
Lee, M.	2021	Title of Paper 9	Key finding nine	Proposed Model I	Empirical evidence
Wilson et al.	2019	Title of Paper 10	Key finding ten	Proposed Model J	Policy implications
Jackson, T.	2022	Title of Paper 11	Key finding eleven	Proposed Model K	Case study analysis
Taylor, S.	2016	Title of Paper 12	Key finding twelve	Proposed Model L	Qualitative data focus
Moore & Harris	2020	Title of Paper 13	Key finding thirteen	Proposed Model M	Statistical significance
Martin, A.	2018	Title of Paper 14	Key finding fourteen	Proposed Model N	Cross-sectional design
Qin and Zhang (2024)	2023	Boosting generalization of fine-tuning BERT for fake news detection	Fine-tuning using an adversarial approach can substantially improve generalisation of models	Feature Regularization-based Adversarial Fine-tuning (FGM-FRAT)	Improvements in performance of between 0.32 and 0.03 (f1-score) relative to base BERT model
King, D.	2019	Title of Paper 16	Key finding sixteen	Proposed Model P	Meta-analysis source
Wright, E.	2021	Title of Paper 17	Key finding seventeen	Proposed Model Q	Unique dataset used
Scott, B.	2020	Title of Paper 18	Key finding eighteen	Proposed Model R	Theoretical framework
Green, C.	2022	Title of Paper 19	Key finding nineteen	Proposed Model S	Experimental setup
Baker et al.	2017	Title of Paper 20	Key finding twenty	Proposed Model T	Generalizability comment

Research Methodology

Datasets

WELFake Dataset

The WELFake dataset was downloaded from Kaggle, and contains 72134 rows of data. The class balance was relatively balanced, with the real news class having an excess of 2000 rows as can be referred to in figure 5. As for features, the dataset came with four columns, an unnamed ID column, title (of the news article), text (from the news article) and the label (real/fake).

H

Figure 4
Enter Caption

Data Understanding

Data Pre-processing

Modelling

Evaluation

Theoretical Framework

models

Implementation Plan

Experiments

Preliminary Results

Gantt chart for FYP2

Conclusion

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